IMEN574 / IDSC574 · PROGRAMMING FOR AX

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Language Models

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Where We Are

01 Programming with AI

Lectures 2–4

02 Python and Classical Machine Learning

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

Outline: Language Models

01 **Contextual Tokens**

How tokens read context

02 **Training and Generation**

How objectives shape outputs

03 **Adapting and Verifying**

How evidence enters answers

01 · Contextual Tokens

01 **Contextual Tokens**
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Why Model Sequences

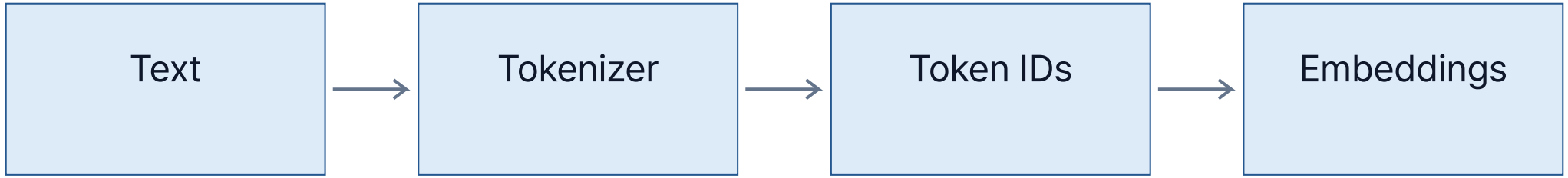
The same token can mean different things in context

Context A	river	bank
Context B	savings	bank

Context changes the information carried by a token

Tokens and Vocabulary

The model predicts token IDs rather than whole ideas



Text becomes an indexed sequence

Worked Example: Token Boundaries

Trace the inputs before execution

```
Text: the cats sat
Greedy vocabulary: cats, cat, sat, the, SPACE
Choose the longest listed token at each position; fallback to one character.
```

Demo: Token Boundaries

Executable educational calculation; no pretrained inference

Compare token rules for the cats sat; edit cats to cat and tokenize again

Text

```
the cats sat
```

Tokenize Rule greedy

Token count = 5

[the] [␣] [cats] [␣] [sat]

Greedy vocabulary: cats, cat, sat, the, SPACE

Token lengths: 3, 1, 4, 1, 3.
Questions: Does “cat” equal “cats”? Can another vocabulary change the sequence?

Answer: Token Boundaries

Same inputs as the initial demo state

```
Greedy tokens [the][SPACE][cats][SPACE][sat], count 5
```

Failure: Token Boundaries

Identify the assumption that breaks

```
This fixed greedy tokenizer is educational, not BERT WordPiece
actual tokenizer IDs depend on model vocabulary
```

Embedding Lookup

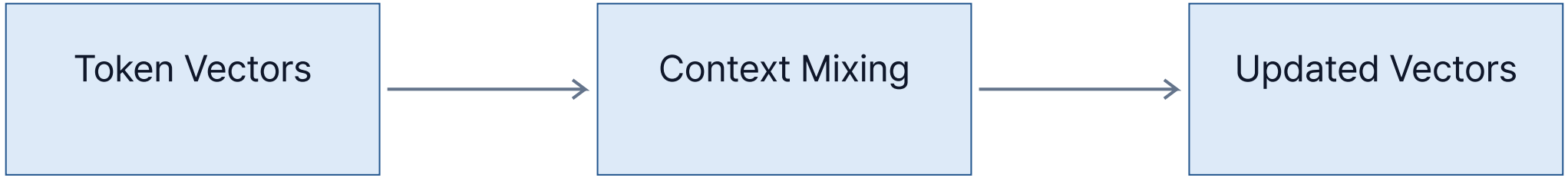
Token IDs select trainable vectors

	d1	d2
ID 0	0.2	0.4
ID 1	0.7	-0.1
ID 2	-0.3	0.9

An ID selects a row in the embedding table

Contextual Representations

A token vector changes after it reads other tokens



Context mixing changes the vector at each position

Demo: Contextual Token Features

MiniLM-L6 replaces BERT-base for contextual token features; 384 hidden dimensions

Run | Code | Data

Original model in Colab ↗

Pretrained model · weights ≈ 23 MB · Original: BERT-base

Input JSON

```
{
  "texts": ["The bank approved
the loan.", "The boat reached the
bank."],
  "word": "bank"
}
```

Run here | Stop | Reset

```
Token: bank
Cosine across contexts: 0.8067
Hidden dimensions: 384

First eight dimensions
A [0.3078, 0.0987, -0.4505, -0.1030, -0.2322, -0.6004, 0.6011, 0.0598]
B [0.1971, 0.5408, -0.9506, -0.7329, -0.5229, -0.3183, 0.4098, 0.3436]
```

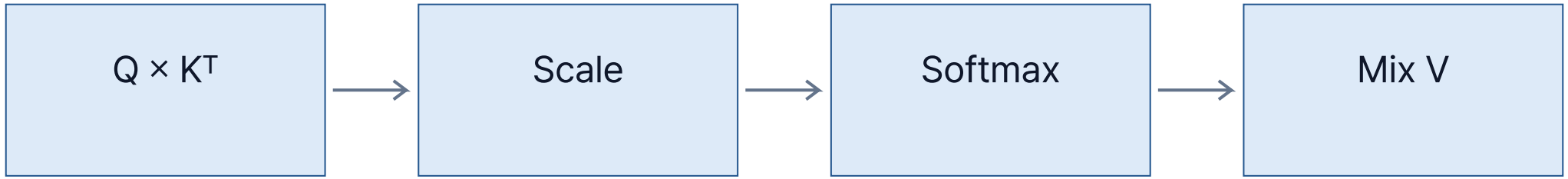
Recorded browser run · Run to recompute

Keep the target token fixed and change only its context. Does its vector stay fixed?

Scaled Dot-Product Attention

Queries choose a weighted mixture of values

$$\text{Attention}(Q,K,V) = \text{softmax}(QK^T / \sqrt{d_k})V$$



Queries, Keys and Values

One input can supply three learned projections

	Query	Key	Value
Token 1	q1	k1	v1
Token 2	q2	k2	v2

Self-attention projects one sequence three ways

Attention Scores

Each query has a distribution across key positions

	key 1	key 2	key 3
query 1	0.7	0.2	0.1
query 2	0.1	0.8	0.1
query 3	0.2	0.3	0.5

Each row defines its own weighted mixture

Worked Example: Causal Attention

Trace the inputs before execution

```
Q = K = [[1,0],[0,1],[1,1]]
V = [[1,0],[0,2],[3,1]]; query index0
Scores = QKᵀ / √2; disallow key positions > query position.
```

Demo: Causal Attention

Executable educational calculation; no pretrained inference

Set future V to 99; compare the output with and without a causal mask

Query token 0 Mask Causal Future V[2,0] 3

Q = K = [[1,0], [0,1], [1,1]]

P(key 0)=1.000	P(key 1)=0.000	P(key 2)=0.000

softmax(QK^T / √2 + mask) V = [1.000, 0.000]

Raw scores: [0.707, 0.000, 0.707].
Questions: Can token 0 see token 2? Does changing future V alter the causal result?

Answer: Causal Attention

Same inputs as the initial demo state

```
Raw scores [.707107,0,.707107]
causal weights [1,0,0]
output [1,0]
```

Failure: Causal Attention

Identify the assumption that breaks

```
Mask future positions before softmax
changing only values must not change attention weights
```

Attention Masks

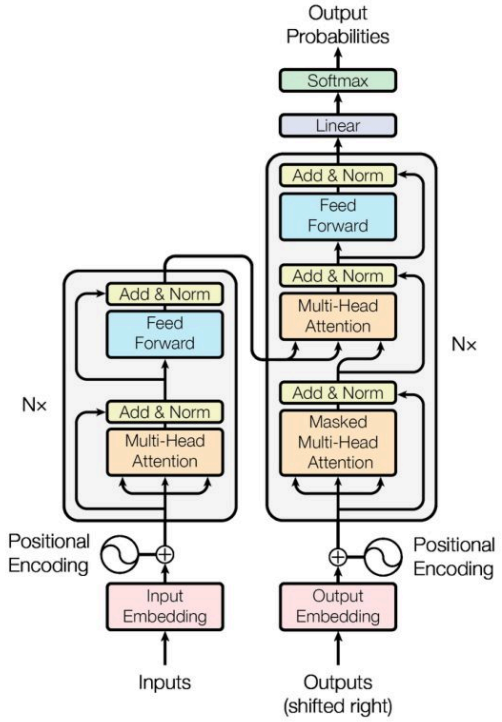
A mask excludes disallowed information paths

	key 1	key 2	key 3
position 1	1	0	0
position 2	1	1	0
position 3	1	1	1

Causal masking blocks future token access

Transformer Block

Attention is one component inside a residual block



The block combines mixing and per-token transformations

https://arxiv.org/abs/1706.03762

02 · Training and Generation

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How evidence enters answers

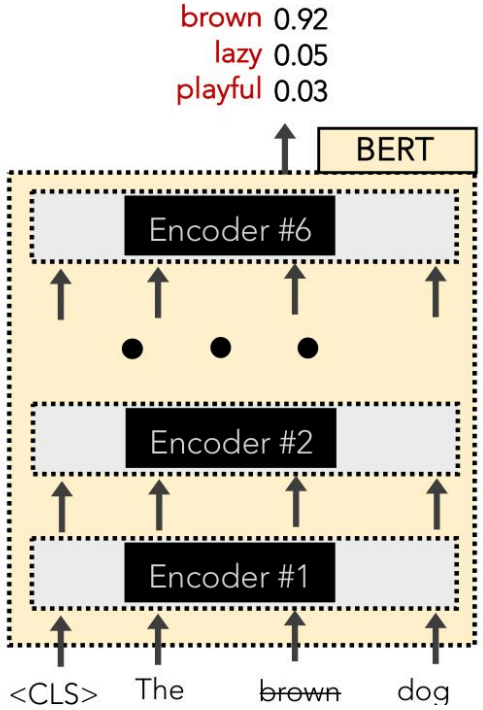
Why Two Training Objectives

Reading all tokens and predicting the next token need different masks

Objective	Visible context	Use
Masked token	Both directions	Representation
Next token	Prefix only	Generation

Masked Language Modeling

The model predicts hidden tokens from surrounding context



Both sides can inform a masked position

https://arxiv.org/abs/1810.04805

Demo: BERT Mask Filling

Same BERT-base model, quantized for the browser

Run | Code | Data

Original model in Colab ↗

Pretrained model · weights ≈ 111 MB · Original: BERT-base

Input JSON

```
{
  "text": "The capital of France
is [MASK].",
  "topK": 5
}
```

Run here | Stop | Reset

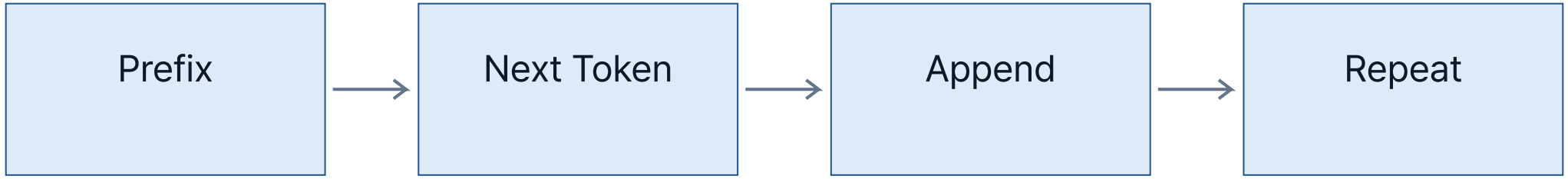
```
paris: 0.3254
lille: 0.0922
lyon: 0.0504
marseille: 0.0476
tours: 0.0385
```

Recorded browser run · Run to recompute

Change the context around [MASK]. Does a high token score establish a fact?

Autoregressive Factorization

A sequence probability is a product of conditional predictions



Each generated token becomes part of the next context

Next-Token Loss

Training compares each prefix with the observed next token

$$L = -\Sigma_t \log p(x_t \mid x_{<t})$$

Shifted Inputs and Targets

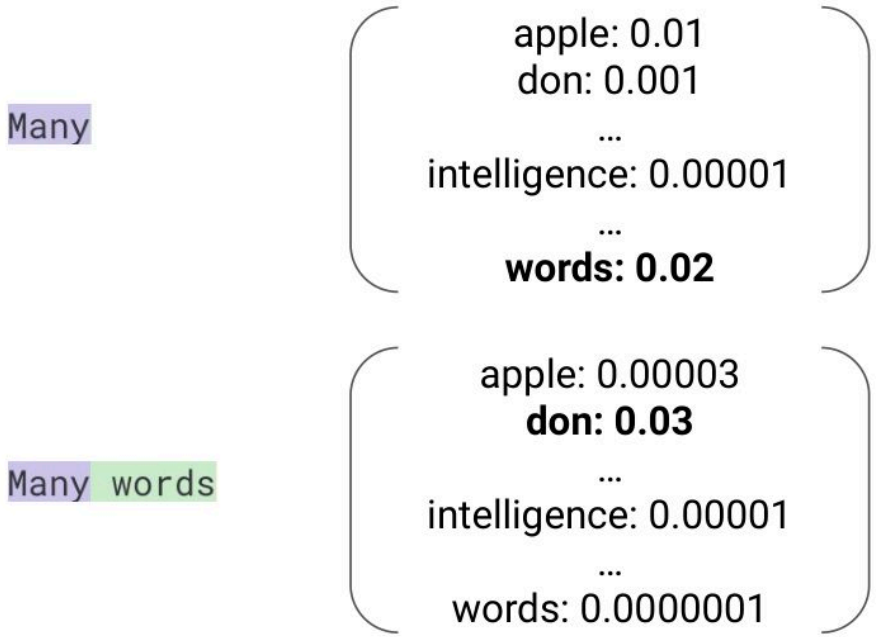
Every observed transition supplies a training example

	t1	t2	t3
Inputs	The	cat	sat
Targets	cat	sat	EOS

Targets are offset by one position

Token Generation

The next-token distribution grows one prefix at a time



A sampled token extends the prefix

Decoding Policy

The same logits can produce different output sequences

Policy	Rule
Greedy	Choose the largest logit
Sampling	Draw from probabilities
Top-p	Restrict cumulative mass

Demo: GPT-2 Generation

Same GPT-2 architecture, quantized for the browser; greedy decoding

Run | Code | Data

Original model in Colab ↗

Pretrained model · weights ≈ 128 MB · Original: GPT-2

Input JSON

```
{
  "text": "The goal of science
is",
  "maxTokens": 12
}
```

Run here | Stop | Reset

```
The goal of science is to understand the nature of the universe and to
understand how it
```

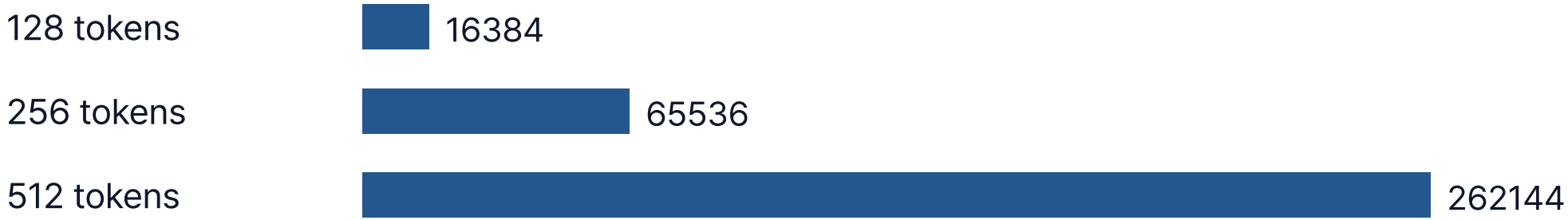
Recorded browser run · Run to recompute

Change the prefix. Does fluent continuation imply instruction following?

https://huggingface.co/openai-community/gpt2

Context Length and Cost

Longer context expands the token interaction matrix

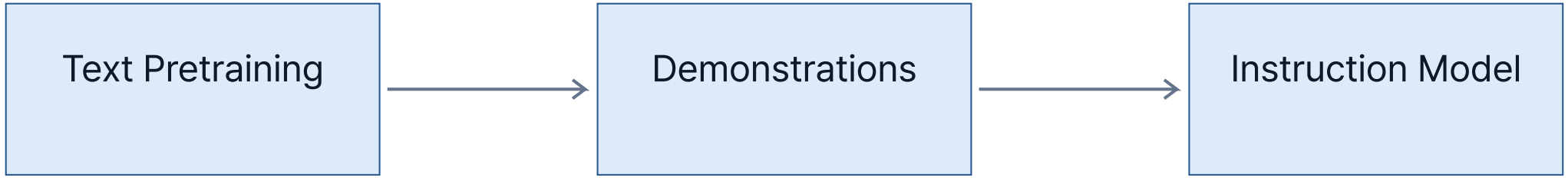


Dense attention score entries

Doubling sequence length quadruples score entries

Pretraining and Instruction Tuning

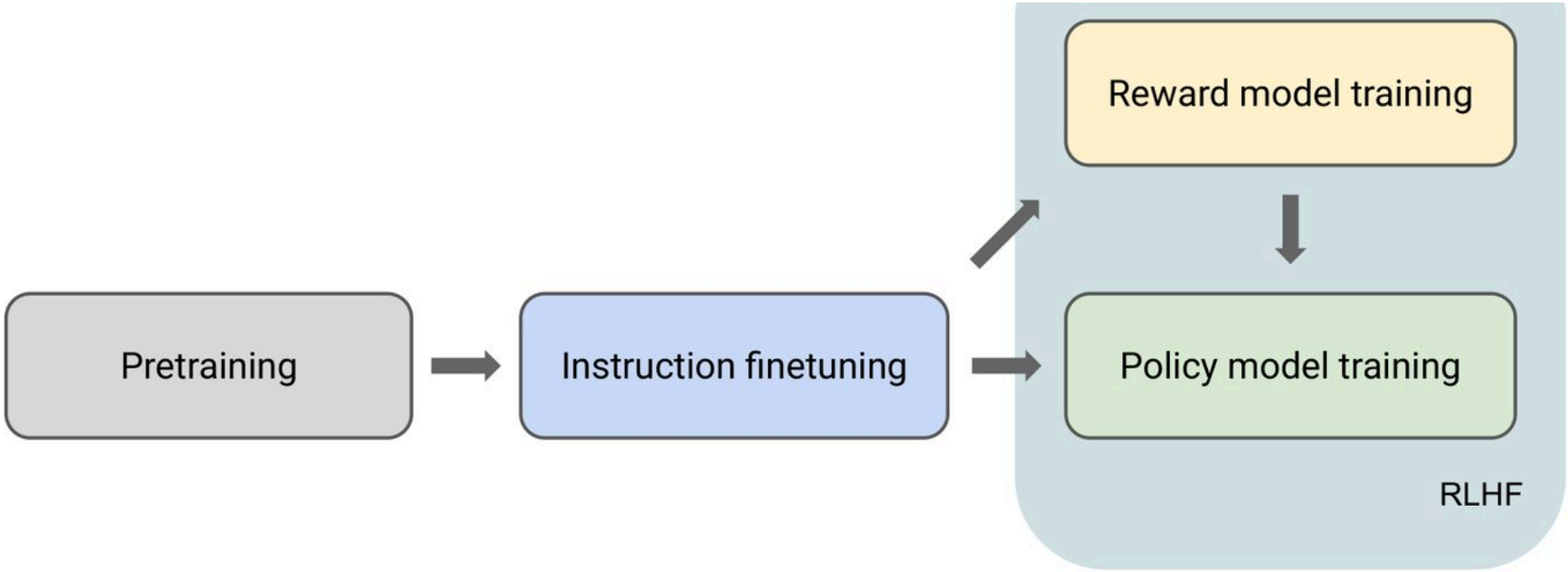
Following a request is trained after learning broad text patterns



Demonstrations teach the model a response format

Human Feedback Pipeline

Preference signals can further shape response behavior



Preference training changes which responses are favored

https://arxiv.org/abs/2203.02155

Caveat: Fluent Errors

Language likelihood does not certify factual truth

Item	Field 1	Field 2
Surface	Plausible wording	high
Evidence	Source support	absent

A likely continuation can lack supporting evidence

03 · Adapting and Verifying

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How evidence enters answers

Why Adaptation Methods Differ

A model may need knowledge, behavior or examples

Need	Mechanism
New document evidence	Retrieval
A response format	Context examples
Persistent adaptation	Parameter updates

In-Context Learning

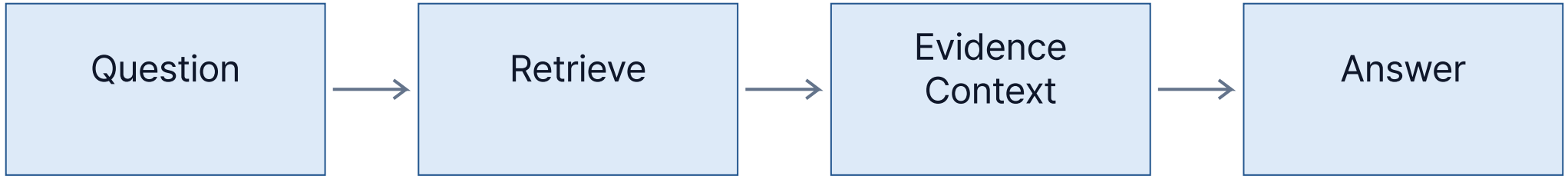
Examples change the context while weights stay fixed

Item	Field 1	Field 2
Examples	great → positive	poor → negative
Query	fine → ?	weights unchanged

Context examples change the conditional prediction

Retrieval-Augmented Generation

Retrieved text supplies evidence for a current question



Evidence enters through the prompt context

Retrieval Ranking

The chosen documents determine the available evidence



Illustrative retrieval similarity

High similarity is a ranking signal, not proof

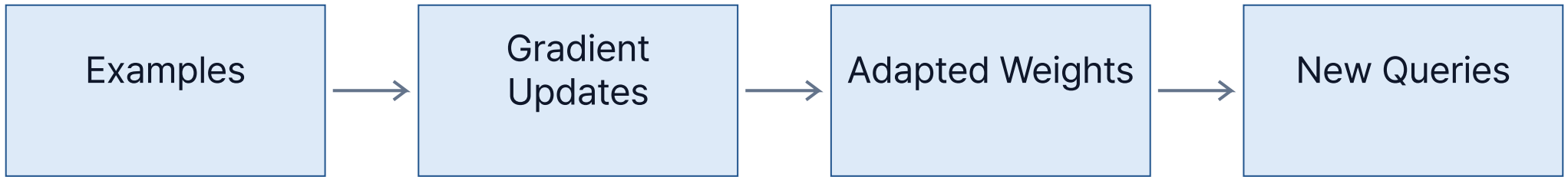
Grounded Answer Contract

An answer must retain links to the supporting passages

Claim	Evidence ID
Valve opened at 10:05	Log 17
Pressure fell later	Sensor 9

Fine-Tuning Scope

Parameter updates persist beyond a single prompt



Fine-tuning changes the stored model parameters

Structured Output Validation

Parseable output still needs semantic checks

```
answer = {"count": 3, "unit": "items"}
assert isinstance(answer["count"], int)
assert answer["count"] >= 0
assert answer["unit"] == "items"
```

Prompt Evaluation Set

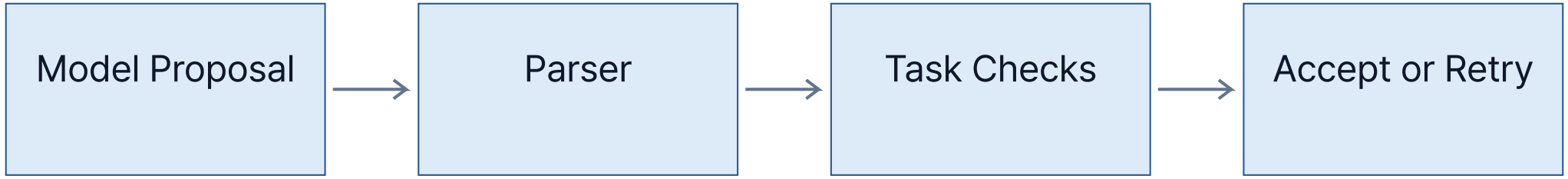
Keep examples that expose known failure modes

	Prompt A	Prompt B	Prompt C
Simple	1	1	0
Ambiguous	1	0	1
Adversarial	0	1	1

Different prompts fail on different input slices

Model and Harness Boundary

Generation supplies a proposal while software checks the result



A software check decides whether to use a proposal

Language Model Audit

Record context, weights and decoding as separate choices

Component	Record
Context	Prompt and retrieved passages
Weights	Model and adaptation
Decoding	Sampling and stop rules

Takeaway: Language Models

01 A language model predicts conditioned token sequences

02 Context, weights and decoding are different controls

03 Fluent output still requires task-specific evidence